

Your Name

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| Page  | Total Points | Score |
|-------|--------------|-------|
| 1     | 16           |       |
| 2     | 12           |       |
| 3     | 6            |       |
| 4     | 6            |       |
| 5     | 4            |       |
| 6     | 10           |       |
| 7     | 10           |       |
| 8     | 8            |       |
| 9     | 8            |       |
| 10    | 10           |       |
| 11    | 10           |       |
| Total | 100          |       |

- You will have 2 hours to complete the test.
- You may bring in 1 page of notes.
- No calculators, cell phones, or computers allowed.

1. (16 points) Evaluate the following integrals.

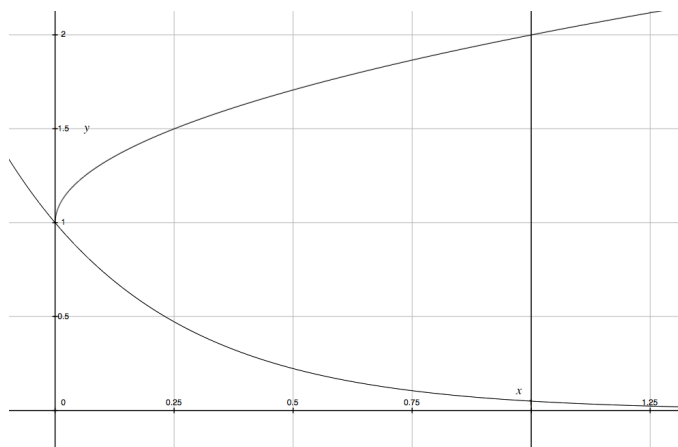
(a)  $\int \sin^3(2x) \cos^2(2x) dx$

(b)  $\int \frac{x-8}{(2x-1)(x-3)} dx$

(c)  $\int \arctan\left(\frac{x}{2}\right) dx$

(d)  $\int \frac{x^2}{(4-x^2)^{1/2}} dx$

2. (12 points) Let  $R$  be the region bounded by the graph of  $f(x) = 1 + \sqrt{x}$ ,  $g(x) = e^{-3x}$  and the vertical line  $x = 1$ , whose graphs are given below.



- (a) Set up, but do not solve, an integral that gives the area of  $R$ .
- (b) Set up, but do not solve, an integral that finds the volume of the solid when  $R$  is rotated about the  $x$ -axis.
- (c) Set up, but do not solve, an integral that finds the volume of the solid when  $R$  is rotated about the  $y$ -axis.

3. (6 points) Let  $a_n = \ln \left( \frac{2n^2 + 1}{3n^2 + 4} \right)$ .

(a) Determine whether the sequence  $a_n$  converges or diverges. If it is convergent determine what it converges to.

(b) Determine whether the series  $\sum_{n=1}^{\infty} a_n$  converges or diverges. Show your work.

4. (6 points) Find the sum of the following series exactly.

a)  $\sum_{n=1}^{\infty} (-3)^{n+1} 5^{-n}$

b)  $\sum_{n=0}^{\infty} \frac{(-1)^n}{2^n n!}$

5. (4 points) Find the Taylor series for the function  $f(x) = e^{3x}$  centered at the point  $a = 5$ . Give your answer in summation notation.

6. (10 points) Use the integral test to determine whether  $\sum_{n=1}^{\infty} \frac{2}{n \ln(n)}$  converges or diverges.

7. (10 points) Determine whether the following series converge or diverge. You must clearly explain your reasoning.

(a)  $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{2n+3}$

(b)  $\sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{n}}{2n+3}$

- (c) Using your answers to (a) and (b), determine whether the series  $\sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{n}}{2n+3}$  is conditionally convergent or absolutely convergent. Explain your reasoning.

8. (8 points) Find the radius of convergence and the interval of convergence of the following series.

(a)  $\sum_{n=1}^{\infty} n!(2x-1)^n$

(b)  $\sum_{n=1}^{\infty} \frac{(x-a)^n}{nb^n}$ , where  $a$  and  $b$  are positive constants.



9. (8 points) Let  $\mathcal{R}$  be the region bounded by  $y = e^x$  and  $y = 0$ ,  $0 \leq x \leq 2$ .

(a) Sketch the region and find the area of  $\mathcal{R}$ .

(b) Find the centroid of the the region  $\mathcal{R}$ .

10. (10 points) Consider  $x = t^2 + 1$ ,  $y = e^{2t} - 1$ .

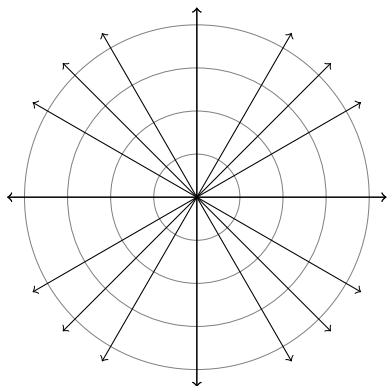
(a) Find  $\frac{dy}{dx}$ .

(b) Is the curve increasing or decreasing when  $t = 2$ ?

(c) Determine the location of any vertical tangents. If none exist, explain why.

11. (10 points) Consider the curve  $r = 1 + 2 \cos \theta$ .

(a) Sketch the curve  $r = 1 + 2 \cos \theta$ .



(b) Find the area enclosed by the inner loop.

**Extra Credit** Let  $f(x) = \sum_{n=0}^{\infty} \binom{\frac{1}{2}}{n} \frac{x^n}{3^n}$ .

1. Determine the first 4 terms in the series. Simplify the coefficients.

2. The series converges when  $x = 2$ . Determine the sum.